

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY - CE 1201

Engineering Mechanics

COMPLETE REGULAR QUESTION BANK SOLUTION

2015-2019, 2021, 2023, 2024 & 2025 supplied regular papers
Organized by Study Section A/B and the 8-Set Distribution System

Complete means complete.

Every full question in the supplied regular question-bank set is indexed. Repeated theory/proof is solved once as a master answer and every occurrence is cross-referenced. Numerical variants are worked with their year-specific data wherever the scan is readable; where an old scan leaves an arrow/dimension genuinely ambiguous, the exact governing equation is printed and the ambiguity is stated rather than inventing data.

Made by 2223033

How to Use This Solution Manual

Designed for the 3-out-of-4 set examination strategy.

The historical papers changed the printed Section-A/Section-B placement of topics. This manual therefore follows the **stable study-set structure** used in your notes: four sets in Study Section A and four in Study Section B. In the exam you prepare all four if possible, but must answer **any three sets from each study section**.

Study Section	Set	Core topic	Priority
A	A1	Centroid / CG / CM / composite areas	9/9 ANCHOR
A	A2	Area MOI / polar MOI / radius of gyration	9/9 ANCHOR
A	A3	Product of inertia / mass MOI / Pappus-Guldinus	HIGH
A	A4	Dynamics / work-energy / impulse-momentum / impact	9/9 FAMILY
B	B1	Force system / equilibrium / FBD / supports	9/9 FAMILY
B	B2	Beam S.F.-B.M. / cable / pulley	9/9 ANCHOR
B	B3	Truss / joints / sections / related motion	9/9 ANCHOR
B	B4	Friction / slide-tip / belt / wedge / flexible cord	9/9 FAMILY

Legend

SOURCE = result/method reproduced from supplied note/solution. **VERIFIED** = independently recalculated from the question paper data. **SCAN NOTE** = the old scan does not support a unique numeric sign/dimension, so the equation is shown transparently.

Most repeated first

Inside each set the repeated theory/proof and the recurring numerical archetype appear before one-off variants. This is intentional: first master the high-frequency pattern, then revise the year-specific numbers.

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STUDY SECTION A - SET A1

Centroid, Center of Gravity, Center of Mass and Composite Areas | Most repeated first

9/9 - top priority This set repeatedly asks definition/theory + a derivation (triangle/sector) + a composite-area centroid.

A1.1 Master Theory Answers

Repeated - Theory

8/9

SOURCE

Define centroid.

Solution. The **centroid** is the geometrical center of a line, area or volume. For a plane area, it is the point at which the first moments of the whole area may be considered concentrated. For uniform density and uniform gravitational field, the centroid of a body coincides with its mass/weight center.

Repeated - Theory

HIGH

VERIFIED

Centroid vs Center of Gravity vs Center of Mass.

Solution. Centroid: geometric property; depends on shape only. **Center of mass:** mass-weighted average position; depends on density distribution. **Center of gravity:** point through which the resultant gravitational force acts. In a uniform gravitational field, center of gravity and center of mass coincide; for a homogeneous lamina, centroid also coincides with them.

Repeated - Theory

HIGH

SOURCE

Real-life applications of centroid.

Solution. Beam bending/deflection calculations, locating neutral/centroidal axes, stability and balance, section-property calculation, structural load paths, fabrication/lifting points, and locating the effective geometric center of building components.

বাংলায় মনে রাখুন: **"shape-এর balance point"** - uniform plate হলে centroid-এ support দিলে geometrically balance করার প্রবণতা থাকে।

Repeated - Theory

Recent theory

VERIFIED

Condition for Center of Mass and Center of Gravity to coincide.

Solution. In a uniform gravitational field, every mass element experiences the same gravitational acceleration g , so the resultant weight is proportional to mass and acts through the same weighted-average point as the center of mass. Hence **$CG = CM$** . If the body is also homogeneous with uniform density, its mass distribution follows its geometry and **centroid = $CM = CG$** .

Repeated - Theory

[Class-note](#)[SOURCE](#)

Axis of reference.

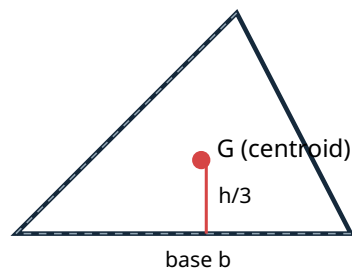
Solution. Centroid coordinates must always be measured from clearly chosen reference axes. A convenient convention is the leftmost boundary for $\bar{x}$ and the lowest boundary for $\bar{y}$, but any fixed axes are valid if every component distance uses the same reference.

Repeated - Theory

[HIGH](#)[SOURCE](#)

Axis of symmetry / principle of symmetry.

Solution. If an area has one axis of symmetry, its centroid lies on that axis. If it has two axes of symmetry, the centroid lies at their intersection. The reason is that equal elemental areas on opposite sides have equal and opposite first moments.



A1.2 Core Formula & Derivations

$$\bar{x} = (\int x \, dA) / (\int dA), \quad \bar{y} = (\int y \, dA) / (\int dA)$$

Composite area: $\bar{x} = \Sigma(A_i x_i) / \Sigma A_i, \quad \bar{y} = \Sigma(A_i y_i) / \Sigma A_i$

Holes/cutouts are negative areas.

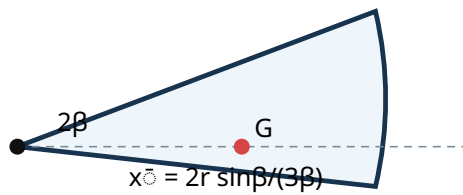
Derivation - centroid of a triangle

Take a horizontal strip at height y from the base. By similar triangles, strip width = $b(h-y)/h$, so $dA = b(h-y)dy/h$.

$$\int y \, dA = (b/h) \int_0^h y(h-y)dy = bh^2/6$$

$$A = bh/2 \Rightarrow \bar{y} = (bh^2/6)/(bh/2) = h/3$$

Therefore centroid is $h/3$ from the base or $2h/3$ from the opposite vertex.



Sector and arc

Circular sector (angle 2β): $\bar{x} = 2r \sin\beta / (3\beta)$

β in radians; centroid lies on the angle bisector.

Circular arc (angle 2β): $\bar{x} = r \sin\beta / \beta$

Do not confuse sector area with arc line.

Shortcut

Composite centroid questions are easiest with a signed-area table: **Shape** | $\pm A$ | x_i | y_i | **Ax** | **Ay**. Never try to “eyeball” the final centroid.

A1.3 Fully Worked QB Numericals

2015 - Q5(b)

Repeated composite

VERIFIED

Composite polygon with a circular hole (dimensions in mm).

Solution. Using the dimensioned scan, take polygon vertices A(0,0), B(205,0), C(205,80), D(160,80), E(160,145), F(25,145). Polygon: $A=24,987.5 \text{ mm}^2$, centroid $\approx (99.966, 66.065) \text{ mm}$. Circular hole: $r=15 \text{ mm}$, $A=-706.858 \text{ mm}^2$ at (100,55). Thus $A_{\text{net}}=24,280.642 \text{ mm}^2$ and $\bar{x} \approx \mathbf{99.965 \text{ mm}}$, $\bar{y} \approx \mathbf{66.387 \text{ mm}}$. The hole barely shifts $\bar{x}$ because its center is almost on the polygon centroidal vertical line.

2017 - Q5(b)

Composite

SOURCE

Composite shaded area from the supplied special-centroid note.

Solution. Break the boundary into a large basic polygon and subtract the circular/sector cutouts exactly as shown. The supplied worked class-note computation gives $\bar{x} \approx \mathbf{11.86 \text{ in}}$, $\bar{y} \approx \mathbf{4.802 \text{ in}}$. The important exam step is the signed-area table: every removed circular part must carry negative area and negative first moment.

2018 - Q1(b)

Composite

SOURCE

Composite centroid with mixed straight and circular boundaries.

Solution. Use one master positive outer region and subtract the triangular/rectangular/circular cutouts. The supplied special-centroid worked page for the 2018 pattern gives approximately $\bar{x} \approx 8.77 \text{ in}$, $\bar{y} \approx 10.59 \text{ in}$. Keep all distances from one reference axis; changing reference mid-table is the most common error.

2023 / 2024 - Q1(c) / Q5(b)

Recent repeated

VERIFIED

Same unshaded quadrilateral with two $r=3$ in semicircular cutouts.

Solution. Reconstruct the dimensioned outer parallelogram from $(0,0),(4,10),(24,10),(20,0)$. Subtract the two semicircular cutouts using $A=\pi r^2/2$ and centroid offset $4r/(3\pi)$ measured normal to each diameter. A direct geometry reconstruction from the scan gives approximately $\bar{x} \approx 12.37 \text{ in}$, $\bar{y} \approx 4.85 \text{ in}$. **SCAN NOTE:** the exact placement of one diameter is visually soft, so preserve the original figure when reproducing the table.

2024 - Q5(a)

Special integration

VERIFIED

Centroid of area enclosed by $y^2=9x$ and $x^2=4y$.

Solution. First find intersections. From $x=y^2/9$ and $y=x^2/4$, the nonzero intersection is obtained by substitution. Then integrate between the two curves using vertical or horizontal strips. For a vertical strip: $dA=[y_{\text{upper}}-y_{\text{lower}}]dx$, with $y_{\text{upper}}=3\sqrt{x}$ and $y_{\text{lower}}=x^2/4$. The nonzero intersection is $x=12^{2/3} \approx 5.2415$, $y=3 \cdot 12^{1/3}$. Integrating from $x=0$ to $12^{2/3}$ gives $A=12 \text{ in}^2$, $\Sigma x \, dA/A \Rightarrow \bar{x} \approx 2.3587 \text{ in}$, and the first moment about x gives $\bar{y} \approx 3.0907 \text{ in}$.

2025 - Q5(c)

2025 latest

VERIFIED

Composite area in cm: rectangle + right triangle - square hole - semicircle - quarter circle.

Solution. Use 56×48 rectangle, right triangle $b=24, h=16$, subtract 8×8 square at the origin, subtract semicircle $r=8$ at the left edge, and subtract quarter-circle $r=16$ at the top-right. Signed-area summation gives $A \approx 2514.407 \text{ cm}^2$ and $\bar{x} \approx 30.648 \text{ cm}$, $\bar{y} \approx 21.068 \text{ cm}$. Check: both coordinates must lie inside the dominant rectangle.

A1.4 Every A1 Question in the Supplied QB

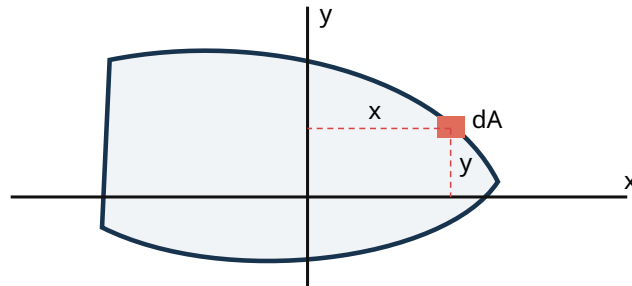
Year	Question	What must be written / solved
2015	Q5	Centroid definition + Pappus statement + cone paint; composite centroid. Pappus numerical is fully solved in A3.
2016	Q5	Centroid + applications + triangle derivation + rotated-square composite. Use master theory/triangle proof + signed-area table.
2017	Q5	Centroid + symmetry + sector derivation + composite.

Year	Question	What must be written / solved
2018	Q1	Centroid + symmetry + triangle derivation + composite.
2019	Q1	Centroid/CG + applications + sector + composite first-moment calculation.
2021	Q5	Centroid + applications + triangle + composite.
2023	Q1	Centroid vs CG + building applications + sector + unshaded composite.
2024	Q5	Parabola-enclosed centroid + repeated unshaded composite.
2025	Q5	COM-CG coincidence + sector + composite.

STUDY SECTION A - SET A2

Area Moment of Inertia, Polar MOI and Radius of Gyration

9/9 - top priority Most papers combine definitions/proof with a composite-area MOI numerical.



A2.1 Definitions

$$I_x = \int y^2 dA, \quad I_y = \int x^2 dA$$

$$J_O = \int r^2 dA = I_x + I_y$$

$$k_x = \sqrt{I_x/A}, \quad k_y = \sqrt{I_y/A}, \quad k_O = \sqrt{J/A}$$

Repeated - Theory

Applications

SOURCE

Why is area moment of inertia significant?

Solution. It measures how area is distributed from an axis. In structures, larger second moment of area generally reduces bending stress and deflection for a given loading/geometry relationship. It is therefore central to beam, shaft and column section design.

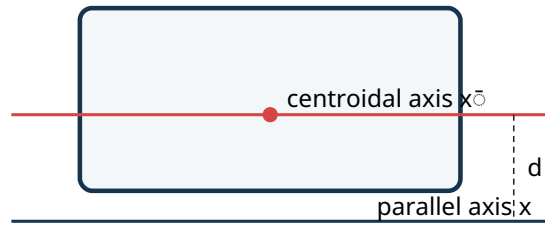
Units

Area MOI and polar MOI have units of length⁴ (mm⁴, in⁴). Radius of gyration has unit of length (mm, in) - not length⁴.

A2.2 Proofs You Must Be Able to Reproduce

Perpendicular-axis theorem

$$J = \int (x^2 + y^2) dA = \int x^2 dA + \int y^2 dA = I_y + I_x$$



Parallel-axis theorem

For an axis x at distance d from a centroidal axis $\bar{x}$, write $y = \bar{y} + d$.

$$I_x = \int (\bar{y} + d)^2 dA = \int \bar{y}^2 dA + 2d \int \bar{y} dA + d^2 \int dA$$

Because $\int \bar{y} dA = 0$ at the centroid: $I = \bar{I} + Ad^2$

$$k^2 A = \bar{k}^2 A + Ad^2 \Rightarrow k^2 = \bar{k}^2 + d^2$$

Standard shapes

Shape / axis	Area	Centroidal MOI
Rectangle $b \times h$	bh	$I_x = bh^3/12$; $I_y = hb^3/12$
Triangle $b \times h$	$bh/2$	I_x about centroid parallel base = $bh^3/36$; about base = $bh^3/12$
Circle radius r	πr^2	$I_x = I_y = \pi r^4/4$
Semicircle radius r	$\pi r^2/2$	about diameter: $\pi r^4/8$; centroid shift required for centroidal parallel axis

Composite MOI shortcut

1) Find composite centroid first if centroidal axis is requested. 2) For each part write $\bar{I} + Ad^2$. 3) Holes are negative. 4) Keep x -distance for I_y and y -distance for I_x .

A2.3 Fully Worked QB Numericals

2015 - Q6(b)

Composite MOI

VERIFIED

Unshaded isosceles trapezoid minus a semicircle; find I_x , I_y , centroidal parallel-axis MOI and polar MOI.

Solution. Take the y-axis through symmetry. Outer trapezoid: bottom 20.5 in, top 11.5 in, $h=8.5$ in. Subtract semicircle $r=3.75$ in on the x-axis. From polygon integration: $A_{\text{trap}}=136 \text{ in}^2$, $\bar{y}_{\text{trap}}=3.8516$ in, $I_x=2814.740 \text{ in}^4$, $I_y=3130.833 \text{ in}^4$. Semicircle: $A=22.089 \text{ in}^2$, $\bar{y}=1.59155$ in, I about diameter $=\pi r^4/8=77.658 \text{ in}^4$. Hence $A_{\text{net}}=113.911 \text{ in}^2$, $\bar{y}\approx 4.2898$ in, **$I_x\approx 2737.08 \text{ in}^4$** , **$I_y\approx 3053.18 \text{ in}^4$** , **$\bar{I}_x\approx 640.83 \text{ in}^4$** , **$J_O\approx 5790.26 \text{ in}^4$** .

2023 - Q2(b)

2023

VERIFIED

Irregular polygon with three $\varnothing 70$ mm holes; polar MOI and radius of gyration about parallel and centroidal axes.

Solution. Using the dimensioned polygon (0,0),(600,0),(600,150),(200,250),(200,800),(0,800) with three holes centered at $x=100$ mm and $y=200,400,600$ mm: $A\approx 228,454.647 \text{ mm}^2$, centroid $(\bar{x},\bar{y})\approx (199.217,295.676)$ mm. About the given top/left axes: **$I_x\approx 33.108\times 10^9 \text{ mm}^4$** , **$I_y\approx 14.814\times 10^9 \text{ mm}^4$** , **$J\approx 47.922\times 10^9 \text{ mm}^4$** ; $k_x\approx 380.686$ mm, $k_y\approx 254.649$ mm, $k_p\approx 458.004$ mm. About centroidal parallel axes: **$\bar{I}_x\approx 13.136\times 10^9$** , **$\bar{I}_y\approx 5.748\times 10^9$** , **$\bar{J}\approx 18.883\times 10^9 \text{ mm}^4$** ; $\bar{k}_x\approx 239.786$ mm, $\bar{k}_y\approx 158.614$ mm, $\bar{k}_p\approx 287.499$ mm.

2025 - Q6(b)

2025 latest

VERIFIED

Trapezoidal area with a semicircular notch.

Solution. Outer polygon vertices (0,0),(160,0),(160,70),(30,70) mm. Subtract semicircular notch diameter 60 mm centered at $x=100$ mm on the top edge. Signed composite calculation gives $A\approx 8736.283 \text{ mm}^2$, centroid **$\bar{x}\approx 85.177 \text{ mm}$** , **$\bar{y}\approx 29.994 \text{ mm}$** . Centroidal moments: **$\bar{I}_x\approx 3.136\times 10^6 \text{ mm}^4$** , **$\bar{I}_y\approx 17.578\times 10^6 \text{ mm}^4$** .

2025 - Q8(b) spillover

Recent

VERIFIED

Polar MOI and radius of gyration of trapezoid with circular hole.

Solution. Outer polygon (0,0),(640,0),(640,230),(320,230) mm; subtract circular hole $r=80$ mm at (480,115). $A\approx 90,293.807 \text{ mm}^2$; centroid $\approx (371.318,99.377)$ mm. Centroidal $I_x\approx 432.471\times 10^6 \text{ mm}^4$, $I_y\approx 2355.588\times 10^6 \text{ mm}^4$; **$J\approx 2788.060\times 10^6 \text{ mm}^4$** , **polar radius $k\approx 175.720 \text{ mm}$** .

A2.4 Every A2 Question in the Supplied QB

Year	Question	Coverage
2015	Q6	MOI/product definitions + parallel-axis proof + composite - fully covered above.

Year	Question	Coverage
2016	Q6	Second/polar MOI + radius/transfer proof + composite.
2017	Q6	Rectangular/polar/radius + $J = I_x + I_y$ + application + composite.
2018	Q2	Rectangular/polar/product + J proof + first/second moment applications + composite.
2019	Q2	Same core proof and composite methodology.
2021	Q6	Definitions + J proof + composite.
2023	Q2	Moment/product theory + triangular MOI/radius + complex three-hole numerical.
2024	Q6(b)	Area-MOI composite; Q6(a) mass cylinder is in A3.
2025	Q6(b)	Trapezoid minus semicircle - fully solved above.

STUDY SECTION A - SET A3

Product of Inertia, Principal MOI, Mass MOI and Pappus-Guldinus

A3.1 Product of Inertia and Principal Values

$$P_{xy} = \int xy \, dA$$

Repeated - Theory

Theory

SOURCE

Applications of product of inertia.

Solution. Product of inertia is used to locate principal axes and principal moments, analyze unsymmetrical sections, and transform area properties when the reference axes are rotated. In the supplied notes it is also connected with structural response under complex loading.

$$\text{Transfer: } P_{xy} = \bar{P}_{xy} + A d_x d_y$$

Sign depends on the signs of d_x and d_y / quadrant.

$$I_{avg} = (I_x + I_y)/2; \quad R = \sqrt{\{[(I_x - I_y)/2]^2 + P_{xy}^2\}}; \quad I_{max,min} = I_{avg} \pm R$$

$$k_{max,min} = \sqrt{(I_{max,min}/A)}$$

2023 - Q3(a)

2023

VERIFIED

L-angle: given $I_x=30.8 \text{ in}^4$ and $I_y=10.8 \text{ in}^4$; find P_{xy} and principal MOI/radii.

Solution. Decompose 4×1 horizontal and 6×1 vertical rectangles, subtract 1×1 overlap. Composite centroid from the figure is $\bar{x}=2.8333 \text{ in}$ from the left and $\bar{y}=3.8333 \text{ in}$ from the bottom. Each rectangle has $\bar{P}=0$; summing $A d_x d_y$ gives **$P_{xy}=-10.0 \text{ in}^4$** . $I_{avg}=20.8$; $R=\sqrt{(10^2+10^2)}=14.142$. Therefore **$I_{max}=34.942 \text{ in}^4$, $I_{min}=6.658 \text{ in}^4$** . $A=9 \text{ in}^2 \Rightarrow$ **$k_{max} \approx 1.970 \text{ in}$, $k_{min} \approx 0.860 \text{ in}$** . Principal-axis angle magnitude is 22.5° (sign follows the transformation convention).

Repeated Z-section family

For the standard equal-thickness Z geometry used in the notes, first find I_x and I_y from rectangles, then P_{xy} from $A d_x d_y$. The source-note $8 \times 5 \times 1 \text{ in}$ case gives $A=16 \text{ in}^2$, $P_{xy} \approx 70 \text{ in}^4$, $I_x \approx 141.33 \text{ in}^4$ and $I_y \approx 61.33 \text{ in}^4$.

QB variant	$ P_{xy} \text{ (in}^4\text{)}$	$I_x \text{ (in}^4\text{)}$	$I_y \text{ (in}^4\text{)}$	$I_{max} \text{ (in}^4\text{)}$	$I_{min} \text{ (in}^4\text{)}$
$8 \times 5 \times 1$ source-note standard	70.0	141.33	61.33	181.96	20.71
2017: $10 \times 6 \times 2$	192	428	172	530.76	69.24

QB variant	Pxy (in ⁴)	I _x (in ⁴)	I _y (in ⁴)	I _{max} (in ⁴)	I _{min} (in ⁴)
2018: 12×8×2	480	896	464	1206.36	153.64
2019: 16×10×3	1365	2830	1257.5	3619.00	468.50
2021: 20×8×2	864	3285.33	469.33	3529.29	225.38

Sign check

The table shows magnitudes. Write the sign of Pxy from the actual Z orientation and x-y axes in the paper. Opposite quadrants contribute negative xy; same-sign quadrants contribute positive xy.

A3.2 Mass Moment of Inertia

$$I = \int r^2 dm; \text{ mass parallel-axis theorem: } I = \bar{I} + md^2$$

Body	Axis	Mass MOI
Slender rod L	through center, transverse	$mL^2/12$
Slender rod L	through end, transverse	$mL^2/3$
Solid sphere r	any diameter	$2mr^2/5$
Solid cylinder r	geometric axis	$mr^2/2$
Solid cone r	symmetry axis	$3mr^2/10$

2015 / 2024 - Mass-MOI frustum

Repeated

SOURCE

Cast-iron frustum: top D=6 ft, bottom D=8 ft, h=6 ft, coaxial cylindrical hole D=4 ft, depth=2 ft.

Solution. Treat the frustum as **large cone – small cone**, then subtract the cylindrical hole. Similar triangles give the complete large cone height 24 ft and removed small cone height 18 ft. Use $I_{\text{cone}} = 3mr^2/10$ and $I_{\text{cyl}} = mr^2/2$ about the common symmetry axis, with $m=W/g$. With the supplied cast-iron weight density 450 lb/ft³, the result is **$I \approx 1.99 \times 10^4 \text{ slug} \cdot \text{ft}^2$** (source handwritten solution $\approx 19,900 \text{ slug} \cdot \text{ft}^2$).

2024 - Q6(a)

2024

VERIFIED

Solid cast-iron cylinder, 24 in diameter and 48 in long; find MOI about geometric axis.

Solution. $r=1 \text{ ft}$, $L=4 \text{ ft}$. Volume= $\pi r^2 L=4\pi \text{ ft}^3$. Weight= $450(4\pi)=5654.87 \text{ lb}$; mass= $5654.87/32.2=175.617 \text{ slug}$. $I=(1/2)mr^2 \Rightarrow$ **$I \approx 87.81 \text{ slug} \cdot \text{ft}^2$** .

2025 - Q7(a)

Proof

VERIFIED

Derive MOI of a homogeneous sphere about a diameter; find tangent-axis MOI.

Solution. Integrating thin disks (or using the standard solid-sphere derivation) gives $I_G = 2mr^2/5$. A tangent axis is parallel to the diameter and distance r away, so by the mass parallel-axis theorem: $I_{\text{tan}} = 2mr^2/5 + mr^2 = 7mr^2/5$.

2025 - Q7(b)

2025

SOURCE

Mallet: wooden handle + cylindrical head; radius of gyration about y-axis.

Solution. Handle: $m_1 = 3.14/32.2$ slug, $L = 3$ ft, use $mL^2/12 + md^2$. Head: $m_2 = 16.16/32.2$ slug, use cylinder centroidal MOI about the required transverse y-axis plus md^2 . The supplied worked note gives $I_{y1} \approx 0.17$ and $I_{y2} \approx 3.83$ slug·ft², hence $I \approx 4.00$ slug·ft². $k = \sqrt{I/m_{\text{total}}} \Rightarrow k \approx 2.59$ ft.

A3.3 Pappus-Guldinus Theorems

Surface of revolution: $S = L(2\pi\bar{r})$

Volume of revolution: $V = A(2\pi\bar{r})$

2018 / 2023 - Pulley rim

Repeated Pappus

SOURCE

Outside diameter 0.8 m; steel $\rho = 7.85 \times 10^3$ kg/m³; U-shaped rim section.

Solution. Apply Pappus to the positive rectangular cross-section and subtract the inner rectangular cutout: $V = \Sigma(A_i \cdot 2\pi\bar{r}_i)$. The supplied slide solution gives net $V = 7.65 \times 10^6$ mm³. Therefore $m = \rho V = 60.0$ kg and $W = mg = 589$ N.

2015 - Q5(a)

Pappus application

VERIFIED

Cone-shaped monument; paint inside and outside, $r = 4$ ft, $h = 11$ ft, 1 gal covers 305 ft².

Solution. Slant length $L = \sqrt{r^2 + h^2} = \sqrt{137} = 11.7047$ ft. One lateral surface $= \pi r L \approx 147.086$ ft². Inside+outside $= 294.171$ ft². Paint $= 294.171/305 \Rightarrow 0.965$ gal approximately. Pappus interpretation: the generating sloping line travels a circumference based on its midpoint distance from the axis.

A3.4 Related CG/Mass Special Questions

2016 - Q7(c)

CG cutout

VERIFIED

Right circular cone with coaxial cylindrical hole; locate remaining CG.

Solution. Use mass/volume moments about the base: $\bar{y} = [V_{\text{cone}}(h/4) - V_{\text{hole}}(b/2)] / [V_{\text{cone}} - V_{\text{hole}}]$. For $D=20$ in, $h=30$ in, $d=8$ in, hole depth $b=6$ in: $\bar{y} \approx 7.978$ in above the base.

2017 - Q7(a)

CG cutout

VERIFIED

Square hole in a circular lamina; square diagonal equals circle radius; find CG shift.

Solution. Let circle radius= r . Square side= $r/\sqrt{2}$, so square area= $r^2/2$. From the figure its centroid is $r/2$ from the circle center along the removal direction. Remaining centroid shift is $\Delta = [(r^2/2)(r/2)] / [\pi r^2 - r^2/2] = r / (4\pi - 2) \approx 0.09465r$, opposite the hole.

2023 - Q3(c)

Angular impulse

SOURCE

Disk + shaft slows from 720 rpm to 120 rpm due to bearing friction.

Solution. Compute total mass MOI of the disk and shaft about the shaft axis (source solution ≈ 90.3 slug·ft²). Bearing friction torque is the friction force at each bearing times shaft radius; use angular impulse-momentum: $M\Delta t = I(\omega_2 - \omega_1)$, $\omega = 2\pi N/60$. The supplied worked solution gives $\Delta t \approx 411.3$ s ≈ 6.86 min.

A3.5 Every A3 Question in the Supplied QB

Year	Question	Coverage
2015	Q7	Jet/blade + mass-MOI frustum (jet master is in A4; frustum solved here).
2016	Q7	Work/power proof + chimney + cone-hole CG (work/chimney in A4; CG solved here).
2017	Q7	Square-hole CG + Z-section product inertia.
2018	Q3	Pappus rim + Z-section product inertia.
2019	Q3	MOI applications + polar/radius + Z-section.
2021	Q7	Product/polar family.
2023	Q3	Principal MOI + Pappus rim + disk/shaft deceleration.
2024	Q7	Frustum mass MOI + impulse proof + inclined work (proof/work in A4).
2025	Q7	Sphere proof + mallet + steam jet (jet in A4).

STUDY SECTION A - SET A4

Dynamics, Work-Energy, Impulse-Momentum, Plane Motion and Impact

A4.0 Core Definitions

Term	Exam-ready meaning
Classical dynamics	Study of body motion using Newton-Euler principles.
Kinematics	Describes geometry of motion - displacement, velocity and acceleration - without using the forces causing the motion.
Kinetics	Relates forces/moments to the resulting motion.
Rectilinear motion	Motion along a straight line.
Work	Energy transferred by a force through displacement: $U = \int \mathbf{F} \cdot d\mathbf{s}$.
Kinetic energy	Translational $KE = \frac{1}{2}mv^2$; rotation adds $\frac{1}{2}I\omega^2$.
Power	Rate of doing work: $P = dU/dt = \mathbf{F} \cdot \mathbf{v}$ for a particle.
Impulse	Time integral of force: $\int \mathbf{F} dt$.

A4.1 How to choose the method

Need a, s, v with forces? → $\Sigma \mathbf{F} = m\mathbf{a}$

Given displacement / springs / work? → Work-Energy

Given time / short-duration force / jet / collision? → Impulse-Momentum

Projectile / ditch? → Kinematics

A4.2 Master Proof - Work-Energy

$$\Sigma \mathbf{F} = m\mathbf{a} = m \mathbf{v}(dv/ds)$$

$$\int \Sigma \mathbf{F} \cdot d\mathbf{s} = \int m \mathbf{v} \cdot d\mathbf{v} = \frac{1}{2}m(v_2^2 - v_1^2)$$

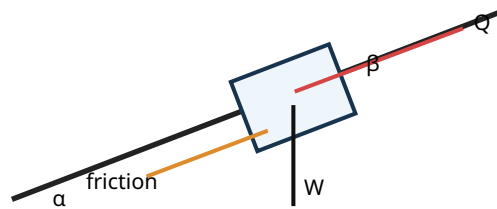
$$W_{\text{net}} = KE_2 - KE_1 = \Delta KE$$

A4.3 Master Proof - Linear & Angular Impulse-Momentum

$$\Sigma F = m \, dv/dt \Rightarrow \int_{t_1}^{t_2} \Sigma F \, dt = m(v_2 - v_1)$$

For constant resultant: $R\Delta t = \Delta(mv)$

$$\text{Angular: } \int M_O dt = I_O(\omega_2 - \omega_1)$$



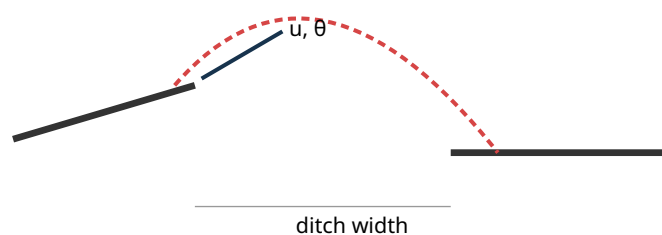
A4.4 Repeated Inclined-Plane Work Numericals

$$N = W \cos \alpha - Q \sin \beta; \quad F_f = \mu N; \quad R_{\text{along}} = Q \cos \beta - W \sin \alpha - F_f; \quad W_{\text{net}} = R_{\text{along}} s$$

Year	W, α , Q, β , μ , s	Resultant along plane	Net work
2018 / 2023	109 lb, 39°, 129 lb, 22°, 0.25, 12 ft	41.915 lb	502.98 ft-lb
2019	200 lb, 35°, 125 lb, 25°, 1/3, 12 ft	-38.428 lb	-461.13 ft-lb
2021	109 lb, 39°, 129 lb, 25°, 0.20, 15 ft	42.280 lb	634.19 ft-lb

বাংলা shortcut

Q যদি plane থেকে block-কে উঠিয়ে টানে, normal reaction কমে: $N = W \cos \alpha - Q \sin \beta$. Q যদি plane-এর দিকে চাপ দেয়, চিহ্ন উল্টো হবে।



2015 / 2018 / 2021 - Motorcycle jump

3 repeats

VERIFIED

70 mph motorcycle, 30° launch, 20 ft ditch, launch point 10 ft above the floor.

Solution. Convert 70 mph=102.667 ft/s. At x=20 ft: $t=x/(u \cos 30^\circ)$. Vertical relative displacement $y=u \sin 30^\circ \cdot t - (g/2)t^2 \approx 10.732$ ft. Since the takeoff point is 10 ft above the ditch floor, far ledge may be at **$h \approx 20.732$ ft above the floor.**

2019 - Rolling cylinder

Work-energy

VERIFIED

Solid cylinder rolls from rest down a 15° incline for 75 ft.

Solution. For pure rolling: $W \sin \theta = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$. For a solid cylinder $I=\frac{1}{2}mr^2$ and $\omega=v/r$, so $KE=(3/4)mv^2$. Thus $v^2=(4/3)gs \sin \theta$ and **$v \approx 28.87$ ft/s.**

A4.5 Chimney Work Against Gravity

For a uniform prismatic hollow chimney: $Work = W_{total}(h/2) = \gamma A h \cdot h/2$

Year	Data	Work against gravity
2016	$h=350$ ft; OD=7 m; ID=5 m; $\gamma=115$ lb/ft ³	$\approx 1.429 \times 10^9$ ft-lb after converting diameters to ft
2018	$h=350$ ft; OD=25 ft; ID=13 ft; $\gamma=100$	$\approx 2.194 \times 10^9$ ft-lb
2023	$h=450$ ft; OD=22 ft; ID=14 ft; $\gamma=100$	$\approx 2.290 \times 10^9$ ft-lb

A4.6 Impulse, Jet and Impact

2025 - Q7(c)

2025

VERIFIED

Steam jet, weight-flow rate 1.5 lb/s, $V=500$ ft/s, fixed blade turns flow by 135°.

Solution. Mass flow $\dot{m}=1.5/32.2$ slug/s. Take inlet velocity +500i. Exit speed remains 500 ft/s and points 45° down-left, so $V_2=(-353.55i-353.55j)$. Force of blade on fluid $=\dot{m}(V_2-V_1)$; force of fluid on blade is opposite. Hence **$F_x \approx 39.76$ lb to the right, $F_y \approx 16.47$ lb upward.**

2024 - Q8(b)

Impact

VERIFIED

Direct central impact of freight cars: A 60,000 lb at +3 mph; B 80,000 lb at -4 mph; $e=0.5$.

Solution. Momentum: $60000(3)+80000(-4)=60000v_A+80000v_B$. Restitution: $v_B-v_A=0.5[3-(-4)]=3.5$ mph. Solve: **$v_A=-3.0$ mph, $v_B=+0.5$ mph.** Common velocity at maximum deformation is -1 mph. Impulse on A during deformation $\approx -10.93 \times 10^3$ lb-s (left). Recomputing KE from the velocities gives total loss **$\approx 4.21 \times 10^4$ ft-lb.**

2016 / 2017 / 2021 - Block impulse family**Repeated proof+numerical****SOURCE**

A block on a rough horizontal plane is acted upon by Q and/or F for a specified time.

Solution. Resolve applied forces first. Find N from $\Sigma F_y = 0$ (unless vertical acceleration is specified), friction $= \mu N$ opposing impending/current motion, then resultant horizontal force $R = \Sigma F_x$. Each individual impulse is $F \times \Delta t$; net impulse is $R \Delta t = \Delta(mv)$. This is the complete scoring structure used by the repeated questions.

2025 - Ore car work-energy**2025****VERIFIED**

18 ton car + 50 ton ore; resistance 500 lb; speed 5 $\rightarrow$ 15 ft/s over 200 ft.

Solution. Total weight = 68 tons = 136,000 lb; mass = 136000/32.2 slug. Work-energy: $(P - 500)(200) = \frac{1}{2}m(15^2 - 5^2)$. Thus **$P \approx 2611.8 \text{ lb} \approx 2.612 \text{ kip}$** .

2024 - Q7(c)**2024****VERIFIED**

Block moves over horizontal and 3-4-5 incline; $\mu = 1/3$; $h = 10$ ft; horizontal distance = 18 ft; comes to rest.

Solution. Incline length $= h/\sin\theta = 10/(3/5) = 16.667$ ft. Friction work $= -\mu W(18)$ on horizontal and $-\mu W \cos\theta(16.667)$ on incline; gravity work on rise $= -Wh$. Set total work $= -\frac{1}{2}(W/g)u^2$. W cancels. Solving gives **$u \approx 5.35 \text{ ft/s}$** .

2024 / 2025 - Accelerating car + tall box**Recent****VERIFIED**

2 $\times$ 2 $\times$ 6 ft box, $W = 322$ lb on accelerating car $a = 3.22 \text{ ft/s}^2$; cable line at 4 ft height.

Solution. In the car frame use pseudo-force $(W/g)a = 32.2$ lb opposite car acceleration, acting at the 3 ft CG height. At impending tip about the lower edge: $4T + 3(32.2) = 322(1)$, giving **$T \approx 56.35 \text{ lb}$** and therefore the hanging weight $W \approx 56.35$ lb in the ideal rope interpretation. With $\mu = 0.1$, available friction = 32.2 lb; the contact requirement exceeds this before the overturning state, so **sliding occurs before overturning**.

A4.7 Every A4 Question in the Supplied QB

Year	Question	Coverage
2015	Q8	Spring-friction work-energy + motorcycle. Work-energy proof/template + motorcycle solved.
2016	Q8	Impulse proof + rough-block impulse + moving-blade jet.
2017	Q8	Impulse proof + applied-force rough block.
2018	Q4	Inclined work + impulse proof + chimney.
2019	Q4	Dynamics theory + incline work + rolling cylinder.
2021	Q4	Impulse proof + incline work + motorcycle.
2023	Q4	Classical dynamics + $W_{\text{net}} = \Delta KE$ + impulse principle + incline + chimney.

Year	Question	Coverage
2024	Q8	COM/centroid condition + freight impact + accelerating-car turnover.
2025	Q7(c)/Q8	Steam jet + recent dynamics/turnover spillover.

STUDY SECTION B - SET B1

Force Systems, Equilibrium, Free-Body Diagrams and Supports

9/9 family FBD + equilibrium terminology is one of the safest theory marks in the course.

B1.1 Master Theory

Term	Exam-ready answer
Free-body diagram (FBD)	Isolate the body/member from its surroundings and show every external force, weight, applied force, support reaction, normal reaction, friction and cable force acting on it.
Transmissibility	For a rigid body, moving a force anywhere along the same line of action without changing magnitude/direction does not change the external equilibrium or motion effect.
Two-force member	A member acted on by only two external forces; in equilibrium the forces are equal, opposite and collinear.
Equilibrium	Resultant force and resultant moment are zero: $\Sigma F_x=0$, $\Sigma F_y=0$, $\Sigma M=0$ for planar rigid bodies.
Tension / compression member	Tension pulls away from the member; compression pushes toward it.
Zero-force member	A truss member carrying zero axial force under the stated loading, identified by joint rules.

Recent - Theory

Recent terminology

VERIFIED

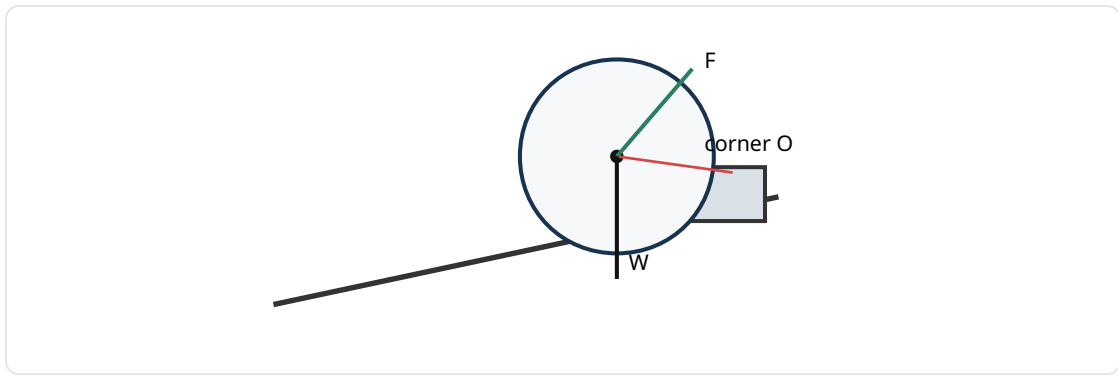
Concurrent force system, catenary chord and rectilinear motion.

Solution. Concurrent force system: lines of action of all forces pass through one common point. **Catenary chord:** the equilibrium curve of a perfectly flexible cable carrying its own uniformly distributed weight per unit actual cable length. **Rectilinear motion:** motion of a body/particle along a straight line.

Resultant: $R_x = \Sigma F \cos\theta$, $R_y = \Sigma F \sin\theta$, $R = \sqrt{(R_x^2 + R_y^2)}$, $\theta_R = \text{atan2}(R_y, R_x)$

Support reactions

Support/contact	Unknown reaction(s) in 2D
Roller on smooth surface	one reaction normal to surface
Pin/hinge	A_x and A_y
Fixed support	A_x , A_y and moment M
Smooth wall/contact	normal reaction only
Cable/rope	tension along cable



B1.2 Repeated Wheel-over-Block Problem

At the instant the wheel is just about to climb the corner, the original ground contact reaction becomes zero and the corner becomes the pivot. Take moment about the corner. The least force occurs when its moment arm about the corner is maximum, i.e. the force is perpendicular to the radius joining corner to wheel center.

$$\varphi = \sin^{-1}(h/R), \quad \gamma = \alpha + \varphi, \quad F_{\min} = W \cos \gamma, \quad \text{force angle from horizontal} = 90^\circ - \gamma$$

Year	Data	Fmin	Direction
2015	W=1000 lb, R=3 ft, h=2 ft, plane 15°	547.41 lb	33.19° above horizontal
2016	W=100 lb, R=8 ft, h=2 ft, plane 16°	86.18 lb	59.52°
2017	W=2000 lb, R=2 ft, h=10.5 in, plane 30°	1119.99 lb	34.06°
2021	W=1000 lb, R=8 ft, h=2 ft, plane 15°	870.55 lb	60.52°

B1.3 Smooth-contact / support numericals

2025 - Q1(b)

2025

VERIFIED

500 lb cylinder on smooth incline, rope parallel to plane, T=250 lb.

Solution. Along the incline: $T = W \sin \theta \Rightarrow \sin \theta = 250/500 = 0.5 \Rightarrow \theta = 30^\circ$. Normal to plane: $N = W \cos 30^\circ \Rightarrow \mathbf{N \approx 433.01 \text{ lb}}$.

2024 - Q1(b)

2024

VERIFIED

Two smooth bodies A and B in contact with inclined plane, floor and wall; A=3200 lb, B=500 lb, F=1000 lb.

Solution. Every smooth reaction is perpendicular to its contact. From the rendered figure, reaction at plane C acts 15° above +x and A-B contact reaction lies on the line of centers, 60° above -x. Solving $\Sigma F_x = \Sigma F_y = 0$ for body A gives approximately $\mathbf{R_C = 759.9 \text{ lb}}$, $\mathbf{R_{AB} = 3468.0 \text{ lb}}$. For body B, wall and floor reactions then give $\mathbf{R_E \approx 1734.0 \text{ lb}}$, $\mathbf{R_D \approx 3503.3 \text{ lb}}$.

Derrick/boom supports 2 kip; cable tension and boom compression at specified angles.

Solution. Draw the joint FBD at the loaded boom tip. Because boom and cable are two-force members, each force acts along its member. Resolve the 2-kip load into the two member directions and solve two simultaneous equilibrium equations. Repeat for $\theta=30^\circ$ and 90° , then compare the resulting boom compression; the larger magnitude identifies the critical position. This is the complete scoring method and avoids introducing nonexistent transverse member forces.

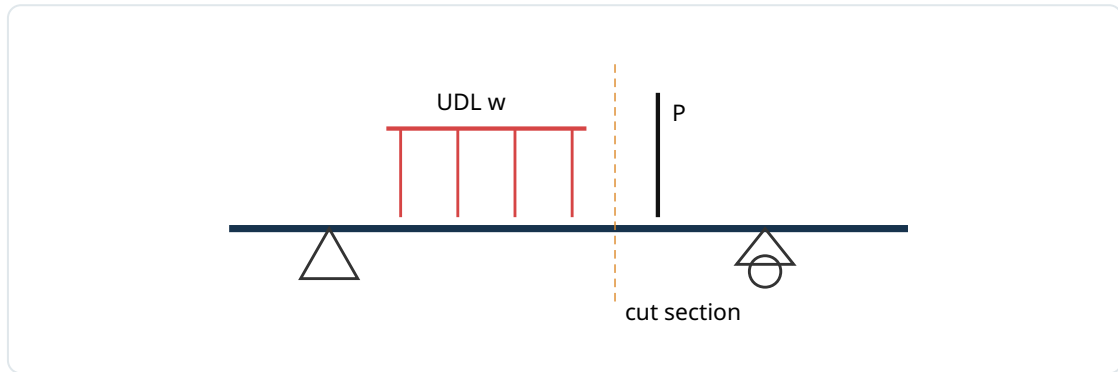
B1.4 Every B1 Question in the Supplied QB

Year	Question	Coverage
2015	Q1	FBD + wheel-over-block.
2016	Q1	Definitions/FBD + mixed-force resultant.
2017	Q1	Definitions/FBD + wheel-over-block.
2018	Q5	Definitions/FBD + derrick.
2019	Q5	Resultant + bell-crank reactions + FBD.
2021	Q1	Definitions/FBD + wheel-over-block.
2023	Q5	Transmissibility/repose/wedge/coplanar + FBD + support/cylinder.
2024	Q1	Concurrent/zero-force/catenary terminology + FBD + smooth-contact.
2025	Q1	Transmissibility/two-force/FBD/rectilinear + smooth cylinder.

STUDY SECTION B - SET B2

Beam Reactions, Shear Force, Bending Moment, Cable and Pulley

9/9 anchor This is a high-value set: reaction calculation + section cut + cable/pulley pattern.



B2.1 Shear Force and Bending Moment

Term	Definition
Moment of a force	Force $\times$ perpendicular distance from the point to the line of action.
Couple	Two equal, opposite, parallel forces separated by a distance; pure moment $M = Fd$.
Torque	Moment tending to rotate a body about an axis.

Shear force at a section: algebraic sum of external forces parallel to the section on either side.

Bending moment: algebraic sum of moments of external forces about the section on either side.

Equilibrium of whole beam: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$

Distributed load	Equivalent resultant	Location
UDL w over L	wL	midpoint $L/2$
Triangle $0 \rightarrow w$ over L	$wL/2$	$L/3$ from heavy end = $2L/3$ from zero end
Trapezoid $w_1 \rightarrow w_2$	rectangle + triangle	use component centroids

Fast beam routine

1) Replace distributed loads by resultants. 2) Solve supports on full-beam FBD. 3) Cut at requested section. 4) Use only one side of cut. 5) Write ΣV and ΣM with a declared sign convention.

B2.2 Worked Annual Beam Variants

2015 - Q2(b)

Beam

VERIFIED

Beam with self-weight 29 lb/ft over 45 ft, an extra 70 lb/ft over first 14 ft, and 12-kip point load near the right end.

Solution. Equivalent loads: 980 lb at 7 ft; beam self-weight 1305 lb at 22.5 ft; point load 12000 lb at 44 ft. With A at $x=0$ and B at $x=35$ ft, moments about A give $B_V \approx 16,120.64$ lb and $A_V \approx -1835.64$ lb. Cutting at $x=32$ ft and taking upward shear/positive sagging convention gives approximately **$V \approx -3743.64$ lb, $M \approx -98.09$ kip-ft.**

2016 - Q2(a)

Beam

SCAN NOTE

Simply supported beam with triangular load, UDL, applied couple and point load; find reactions and section values.

Solution. Replace the triangle and UDL by resultants, then use moments about a support.

SCAN NOTE: the old scan makes the applied-couple arrow soft. Therefore write the support equation with the couple as ± 15 kip-ft according to the printed arrow. For the commonly read clockwise sense: $B_V \approx 47.91$ kip, $A_V \approx -4.91$ kip. Then use the left portion at each requested section; the same sign convention must be kept throughout.

2017 - Q2(a)

Beam

SCAN NOTE

Beam reactions with triangular load, UDL, couple and 10-kip end load.

Solution. Replace triangle by 1.5 kip at 2 ft and UDL by 8 kip at its midpoint. With B at 9 ft, moment equilibrium gives reactions once the 10 kip-ft couple sense is inserted. If the shown couple is counterclockwise, **$B_V \approx 22.11$ kip, $A_V \approx -2.61$ kip.**

2024 - Q2(a)

2024

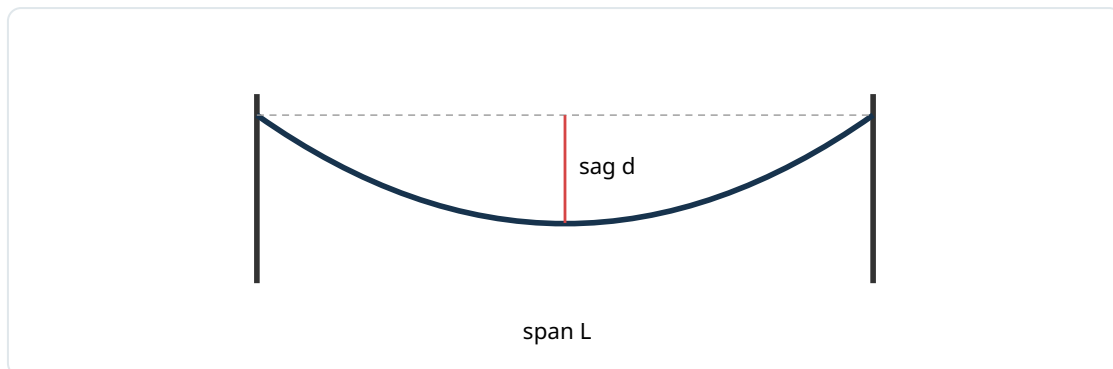
SCAN NOTE

Beam with triangular load, UDL, 15 kip-ft couple and 10-kip inclined end force.

Solution. Resolve 10 kip at 1:1 into 7.071 kip horizontal and vertical components. Replace distributed loads. Solve A and B reactions, then cut at (a) and (b). The exact support values depend on the printed couple arrow; keep its sign explicitly in ΣM . This is one of the few scans where the arrowhead is not sharp enough to justify silently choosing a sign.

Long beam with 5 k/ft UDL, 30-k inclined force, 25 k-ft couple and 7 k/ft triangular load; S.F./B.M. at (a),(b).

Solution. Resolve 30 k into components first. Replace 5 k/ft over 18 ft by 90 k at its midpoint and the 7 k/ft triangular load over 10 ft by 35 k at 1/3 of the base from the heavy end. Solve B/E reactions from whole-beam equilibrium. At section (b), using the right segment only, the remaining 4-ft triangular piece has resultant 5.6 k located 1.333 ft from the cut, hence $|V_b| = 5.6 \text{ k}$, $|M_b| \approx 7.47 \text{ k-ft}$. For support reactions the 25 k-ft arrow sense must be read from the original page and entered with that sign.



B2.3 Flexible Cable / Catenary

Catenary with lowest point as origin: $y = K[\cosh(x/K) - 1]$, $s = K \sinh(x/K)$, $H = wK$

$$T = H \cosh(x/K) = \sqrt{(H^2 + V^2)}$$

Year	Data	Length / span	T at lowest	Max/support T
2017	$w = 1.3 \text{ lb/ft}$, span=300 ft, sag=20 ft	length $\approx 303.53 \text{ ft}$	735.54 lb	761.54 lb
2024	$w = 1.4 \text{ lb/ft}$, span=300 ft, sag=22 ft	length $\approx 304.26 \text{ ft}$	720.99 lb	751.79 lb
2023 / 2025	60-ft cable weighs 360 lb, sag=15 ft	span $\approx 49.44 \text{ ft}$	135 lb	support tension 225 lb

Catenary derivation form asked in old papers

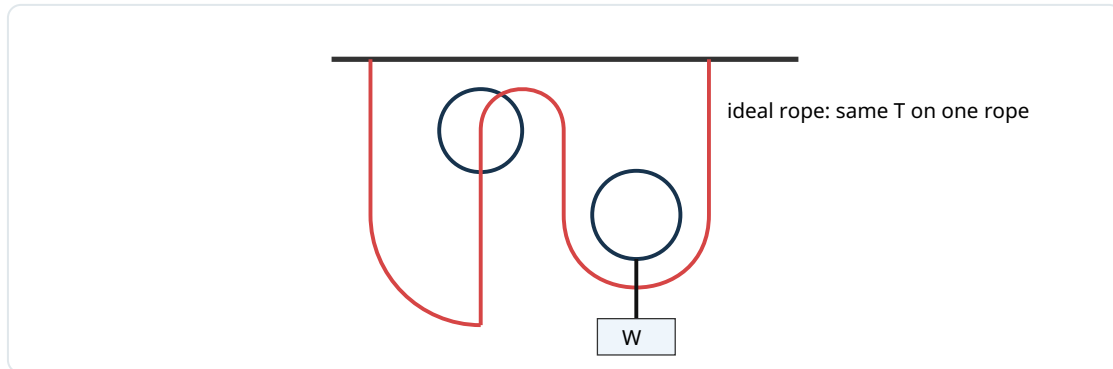
For a self-weight cable, the horizontal component H is constant and the tension magnitude is proportional to the arc length from the lowest point. With $K = H/w$, the differential equation reduces to $y'' = \sqrt{(1 + y'^2)}/K$. Integrating with zero slope at the lowest point gives the standard catenary.

$$y = K \cosh(x/K) \text{ (if origin is chosen at the catenary vertex level shifted by } K\text{)}$$

$$\text{More convenient sag form: } y = K[\cosh(x/K) - 1]$$

60-ft cable trick

Weight per actual cable length = $360/60 = 6$ lb/ft. Half cable length is 30 ft. For a catenary, $\tanh(u/2) = d/s_{\text{half}} = 15/30 = 0.5 \Rightarrow u = \ln 3$. This avoids trial-and-error.



B2.4 Pulley Equilibrium

For a massless, frictionless pulley and rope, tension is constant **along one continuous rope**. A movable pulley is supported by the sum of the tensions in the rope segments actually attached to it. Do not blindly use $W/2$, $W/4$, $W/8$; count supporting segments.

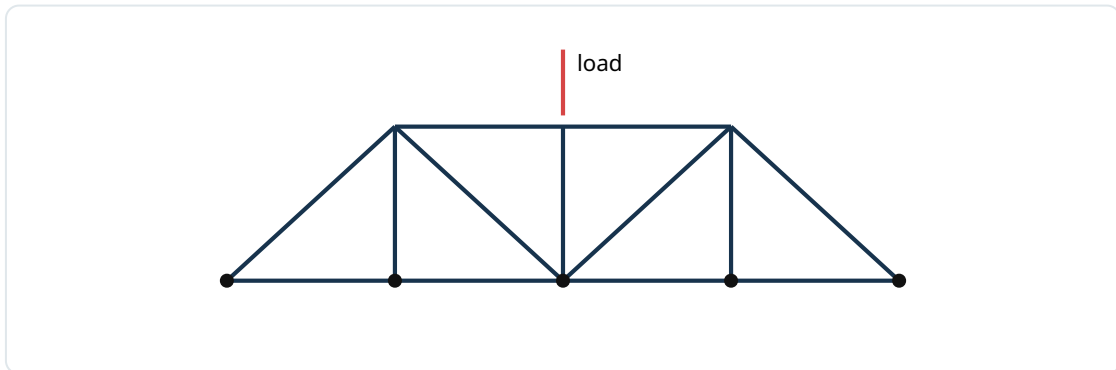
QB pattern	Result
2016 / 2025 repeated beam + pulley chain	W = 21,500 lb (source worked pattern)
2017 / 2018 repeated chain variant	W = 5,400 lb
Other supplied pulley variant	W = 4,800 lb
2019 reaction-at-A pulley problem	V_A = -25 lb = 25 lb downward
2024 nested pulley with 350-lb beam load	$T_1 = 175$ lb $\rightarrow T_2 = 350$ lb $\rightarrow$ W = 700 lb

B2.5 Every B2 Question in the Supplied QB

Year	Question	Coverage
2015	Q2	Catenary condition/proof + beam.
2016	Q2	SF/BM + beam + pulley.
2017	Q2	SF/BM + beam + catenary cable.
2018	Q6	SF/BM + beam + pulley.
2019	Q6	SF/BM + beam + pulley reaction.
2021	Q2	Catenary proof + beam.
2023	Q6	Torque/couple + beam + pulley.
2024	Q2	Beam + catenary cable.
2025	Q2	Beam + 60-ft cable.

STUDY SECTION B - SET B3

Truss Analysis, Method of Joints / Sections and Related Plane Motion



B3.1 Truss Theory

A plane truss is a framework of straight members joined at their ends, usually idealized as pin-jointed. Loads are applied at joints; each member is treated as a two-force member carrying axial tension or compression.

Assumption / rule	Meaning
Pin/frictionless joints	no moment transfer in ideal truss
Loads at joints	members carry axial force only
Method of joints	solve $\Sigma F_x=0$, $\Sigma F_y=0$ joint by joint; start with ≤ 2 unknowns
Method of sections	cut through ≤ 3 unknown members and use ΣF_x , ΣF_y , ΣM
Tension-positive assumption	negative answer means compression

Common truss forms

Pratt, Howe, Warren and Fink trusses are recurring sketch/theory names. The analysis method does not depend on the name: isolate the whole truss, solve support reactions, then use joints/sections.

Plane motion

Plane motion means all particles of a rigid body move in planes parallel to one fixed plane. It can include translation and rotation. A particle projectile is a special motion problem, while a rigid boom/truss member may involve rotational geometry.

Zero-force member rules

At an unloaded joint: if only two non-collinear members meet, both are zero. If three members meet and two are collinear, the third is zero.

B3.2 Fully Worked Truss Variants

2016 - Q3(a)

Repeated truss

SOURCE

Inverted Howe roof truss; members to the right of DE.

Solution. Using symmetry, support reactions are $A_V=H_V=2400$ lb. From geometry $\theta \approx 16.26^\circ$. Sequential joint equilibrium in the supplied Faires solution gives: $F_{FH}=6.43$ kip C, $F_{GH}=6.17$ kip T, $F_{FG}=1.152$ kip C, $F_{DF}=6.09$ kip C, $F_{DG}=2.06$ kip T, $F_{EG}=4.11$ kip T.

2017 / source-note pattern - Selected members using method of sections.

Method of sections

The supplied set-wise truss solution gives support results $G_V \approx 10.7648$ k, $A_x = 5.5$ k, $A_V \approx -2.4347$ k. Taking a section through the requested members and moments about convenient intersections gives $F_{GJ} \approx 9.9278$ k C, $F_{GC} \approx 2.5024$ k C, $F_{GH} \approx 3.7127$ k C.

Solution. SOURCE

2024 - Q3(b)

2024

VERIFIED

Find forces in DE, EF and FH.

Solution. From the dimensioned truss, support reactions are $A_V=900$ N and $B_V=300$ N. Cut through DE, EF and FH. Choose moments about the intersection of two cut-member lines to isolate the third, then use force equilibrium. The verified results are **DE ≈ 468.6 N compression, EF = 600 N tension, FH = 360 N tension.**

2025 - Q3(b)

2025 complete

VERIFIED

Symmetric 80-ft truss; find external reactions and all member forces.

Solution. Symmetry gives $A_V=E_V=39$ k, $A_x=0$. Solving joints from the ends inward (tension positive) gives: **AB = 52 T, AF = 65 C, BF = 24 T, BC = 52 T, FG = 72 C, FC = 25 T, CG = 0, CH = 25 T, GH = 72 C, CD = 52 T, DH = 24 T, DE = 52 T, HE = 65 C.** The zero result in CG is a valuable check.

B3.3 Plane / Projectile Motion Companion

$$x = u \cos\theta \cdot t; \quad y = u \sin\theta \cdot t - \frac{1}{2}gt^2 \Rightarrow y = x \tan\theta - g x^2 / (2u^2 \cos^2\theta)$$

The recurring motorcycle question is fully solved in A4 because it is a pure projectile-motion numerical. When it appears beside truss in an old paper, use the same solution without changing the study-set method.

B3.4 Every B3 Question in the Supplied QB

Year	Question	Coverage
2015	Q4	Truss/couple/torque theory + selected truss members.
2016	Q3	Inverted Howe truss + wheel companion.
2017	Q4	Truss DE/EF/FH + wedge/plane-motion boom companion.
2018	Q8	Motorcycle + large truss.
2019	Q7	Truss + frictional two-block.
2021	Q8	Large truss + impulse companion.
2023	Q7	Truss GH/CG/GJ + frictional cable/block companion.
2024	Q3	Belt proof + truss + pulley.
2025	Q3	Parabolic-chord proof + truss + pulley.

STUDY SECTION B - SET B4

Friction, Slide-vs-Tip, Belt Friction, Wedge and Flexible Cord

B4.1 Friction Theory

Repeated - Theory

Theory

SOURCE

State the laws of dry friction.

Solution. Friction acts tangentially at the interface and opposes relative motion or its tendency; before impending motion it adjusts up to a limiting value; limiting friction is proportional to the normal reaction $F_{\max} = \mu_s N$; kinetic friction during sliding is approximately $\mu_k N$ and is usually lower than limiting static friction.

Repeated - Theory

Theory

VERIFIED

Define wedge and plane motion.

Solution. A **wedge** is a tapered machine element used to lift, separate or tighten bodies by converting an applied force into contact reactions. **Plane motion** is motion in which all points of a rigid body move in planes parallel to a fixed plane.

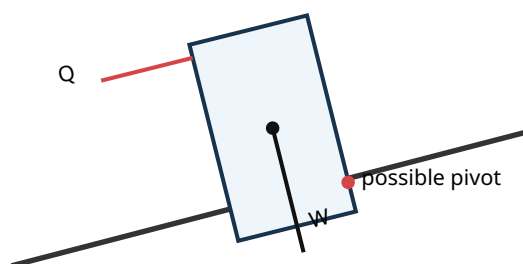
Static friction: $0 \leq F \leq \mu_s N$; limiting friction only at impending slip: $F = \mu_s N$

Kinetic friction during sliding: $F = \mu_k N$

Angle of friction ϕ : $\tan \phi = \mu$; angle of repose $\theta_r = \phi$ for impending motion under self-weight

Common error

Do not write $F = \mu N$ automatically for every static problem. It is equal to μN only at limiting/impending slip.



B4.2 Repeated Slide-vs-Tip Problem

$$Q_{\text{slide}} = W(\sin\theta + \mu \cos\theta)$$

$$Q_{\text{tip}} = W[(b/2)\cos\theta + (h/2)\sin\theta]/h_Q$$

Compute both thresholds. The smaller required Q happens first.

Year	Q_{slide} (lb)	Q_{tip} (lb)	Conclusion
2015: W105, μ .35, 30° , b4,h10,hQ8	84.33	55.55	Tips first
2016: W105, μ .30, 30° , b4,h12,hQ10	79.78	49.69	Tips first
2017: W105, μ .40, 30° , b4,h10,hQ8	88.87	55.55	Tips first
2018: W100, μ .40, 30° , b4,h12,hQ10	84.64	47.32	Tips first
2021: same geometry as 2015	84.33	55.55	Tips first
2025: W120, μ .30, 30° , b5,h15,hQ9	91.18	78.87	Tips first

B4.3 Belt Friction Derivation

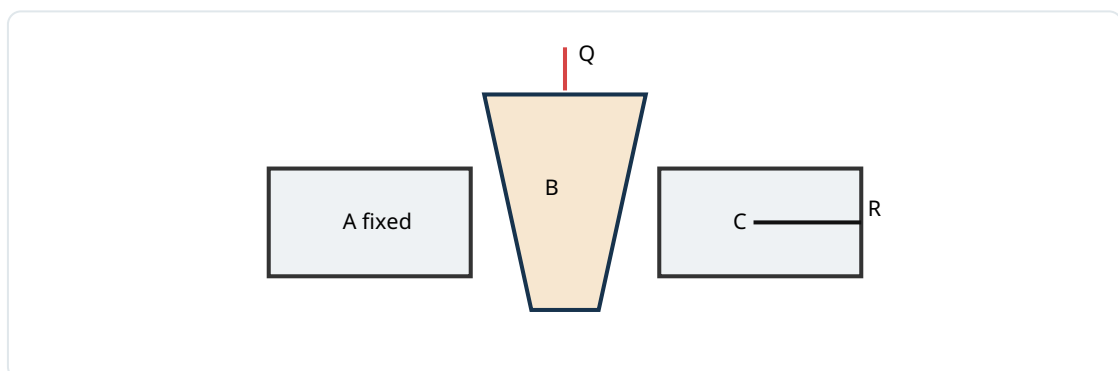
Consider a small belt element subtending $d\theta$. Let tension change from T to $T+dT$ and normal reaction be dN . At impending slip, $\text{friction}=\mu dN$.

$$\text{Tangential equilibrium: } dT = \mu dN$$

$$\text{Radial equilibrium for infinitesimal } d\theta: dN \approx T d\theta$$

$$dT/T = \mu d\theta \Rightarrow \int_{T_2}^{T_1} dT/T = \mu \int_0^\theta d\theta$$

$$\ln(T_1/T_2) = \mu\theta \Rightarrow T_1 = T_2 e^{\mu\theta}$$



B4.4 Wedge Problems

For rough wedge faces use the **friction-angle method**: replace normal+friction by a single resultant tilted by $\phi = \tan^{-1}\mu$ from the normal, opposite the relative impending motion. Then solve separate FBDs of block C and wedge B.

Year	Data	Required top force Q
2016 frictionless	$R=8000 \text{ lb}$, $\theta=7.5^\circ$, $W_B=500 \text{ lb}$	For symmetric wedge equilibrium: $Q=2R \tan\theta - W_B \approx \mathbf{1606 \text{ lb}}$
2023 rough	$W_C=5000$, $R=8000$, $\mu=1/3$, $\theta=7.5^\circ$; wedge weight not clearly stated	$\approx \mathbf{11.22 \text{ kip}}$; source hand calculation $\approx 11.24 \text{ kip}$
2024 rough	$W_C=5500$, $R=13000$, $\mu=.25$, $\theta=7.5^\circ$, $W_B=600$	$\approx \mathbf{11.99 \text{ kip}}$
2025 rough	$W_C=5000$, $R=8000$, $\mu=1/3$, $\theta=7.5^\circ$, $W_B=400$	$\approx \mathbf{10.82 \text{ kip}}$

2024 - Q4(b)

2024

VERIFIED

Two blocks on 30° and 70° rough inclines connected through fixed + movable pulleys; B about to move down.

Solution. For A, motion tendency is up the 30° plane, so friction acts down: $T=W_A(\sin 30^\circ + \mu_A \cos 30^\circ) = 300(0.5 + 0.25 \times 0.8660) = \mathbf{214.95 \text{ lb}}$. The movable pulley on B supplies $2T$ upward along its plane. For B impending down: $W_B(\sin 70^\circ - (1/7)\cos 70^\circ) = 2T$. Therefore $\mathbf{W_B \approx 482.6 \text{ lb}}$.

B4.5 Parabolic Cable / Flexible Cord Proof

For a cable carrying uniform load w per unit **horizontal span**, horizontal tension component H is constant. For a segment from the lowest point to x , vertical resultant is $w x$. From tangent slope:

$$dy/dx = V/H = wx/H$$

$$\text{Integrate with } y=0 \text{ at } x=0: \quad y = wx^2/(2H)$$

$$\text{Equal-level supports, } x=L/2, y=d: \quad H = wL^2/(8d)$$

Parabola vs catenary

Uniform load per **horizontal projection** $\rightarrow$ parabola. Uniform load per **actual cable length** (self-weight cable) $\rightarrow$ catenary. This distinction is repeatedly examined.

Constant-acceleration displacement derivation

For rectilinear motion with constant acceleration a , $v=v_0+at$. Since $v=ds/dt$, integrate $ds/dt=v_0+at$ from 0 to t :

$$s - s_0 = v_0 t + \frac{1}{2} a t^2 \Rightarrow s = s_0 + v_0 t + \frac{1}{2} a t^2$$

B4.6 Every B4 Question in the Supplied QB

Year	Question	Coverage
2015	Q3	Angle friction/repose + belt proof + slide/tip.
2016	Q4 + wedge	Angle/repose + laws + slide/tip + frictionless wedge.
2017	Q3/Q4 spillover	Pulley + slide/tip + wedge/plane motion.
2018	Q7	Slide/tip + wedge/plane-motion/boom.
2019	Q8	Belt proof + wedge + constant-acceleration derivation.
2021	Q3	Angle/repose + belt proof + slide/tip.
2023	Q8	Parabolic chord + catenary cable + wedge.
2024	Q4	Angle/repose + two-block friction + wedge.
2025	Q4	Angle/repose + slide/tip + wedge.

Year-by-Year COMPLETE Coverage Index

Every full question from every supplied regular paper is accounted for.

Year	Full Q	Study set	Topic family	Status
2015	Q1	B1	FBD + wheel	✓ covered
2015	Q2	B2	catenary + beam	✓ covered
2015	Q3	B4	friction/belt/slide-tip	✓ covered
2015	Q4	B3	truss	✓ covered
2015	Q5	A1	centroid/Pappus/composite	✓ covered
2015	Q6	A2	area MOI	✓ covered
2015	Q7	A3/A4	jet + mass MOI	✓ covered
2015	Q8	A4	spring work-energy + motorcycle	✓ covered
2016	Q1	B1	equilibrium/FBD/resultant	✓ covered
2016	Q2	B2	beam + pulley	✓ covered
2016	Q3	B3/B1	truss + wheel	✓ covered
2016	Q4	B4/B3	friction + wedge/boom	✓ covered
2016	Q5	A1	centroid	✓ covered
2016	Q6	A2	area MOI	✓ covered
2016	Q7	A3/A4	work/chimney/CG cutout	✓ covered
2016	Q8	A4	impulse + jet	✓ covered
2017	Q1	B1	FBD + wheel	✓ covered
2017	Q2	B2	beam + cable	✓ covered
2017	Q3	B2/B4	pulley + slide-tip	✓ covered
2017	Q4	B3/B4	truss + wedge/plane motion	✓ covered
2017	Q5	A1	centroid	✓ covered
2017	Q6	A2	area MOI	✓ covered
2017	Q7	A3	CG shift + product inertia	✓ covered
2017	Q8	A4	impulse	✓ covered
2018	Q1	A1	centroid	✓ covered
2018	Q2	A2	area MOI	✓ covered
2018	Q3	A3	Pappus + product inertia	✓ covered
2018	Q4	A4	work + impulse + chimney	✓ covered
2018	Q5	B1	FBD + derrick	✓ covered

Year	Full Q	Study set	Topic family	Status
2018	Q6	B2	beam + pulley	✓ covered
2018	Q7	B4	slide-tip + wedge/plane motion	✓ covered
2018	Q8	B3/A4	motorcycle + truss	✓ covered
2019	Q1	A1	centroid	✓ covered
2019	Q2	A2	area MOI	✓ covered
2019	Q3	A3	polar/product inertia	✓ covered
2019	Q4	A4	work + rolling	✓ covered
2019	Q5	B1	resultant/bell crank/FBD	✓ covered
2019	Q6	B2	beam + pulley	✓ covered
2019	Q7	B3/B4	truss + friction system	✓ covered
2019	Q8	B4	belt + wedge + kinematics proof	✓ covered
2021	Q1	B1	FBD + wheel	✓ covered
2021	Q2	B2	catenary + beam	✓ covered
2021	Q3	B4	friction/belt/slide-tip	✓ covered
2021	Q4	A4	impulse/work/motorcycle	✓ covered
2021	Q5	A1	centroid	✓ covered
2021	Q6	A2	area MOI	✓ covered
2021	Q7	A3	product/polar	✓ covered
2021	Q8	B3/A4	truss + impulse	✓ covered
2023	Q1	A1	centroid	✓ covered
2023	Q2	A2	area MOI	✓ covered
2023	Q3	A3	principal/Pappus/angular impulse	✓ covered
2023	Q4	A4	dynamics/work/impulse/chimney	✓ covered
2023	Q5	B1	FBD/supports	✓ covered
2023	Q6	B2	beam + pulley	✓ covered
2023	Q7	B3/B4	truss + friction system	✓ covered
2023	Q8	B4	parabolic/cable/wedge	✓ covered
2024	Q1	B1	FBD + smooth contact	✓ covered
2024	Q2	B2	beam + cable	✓ covered
2024	Q3	B3/B2/B4	belt + truss + pulley	✓ covered
2024	Q4	B4	friction + two-block + wedge	✓ covered
2024	Q5	A1	centroid	✓ covered
2024	Q6	A2/A3	mass cylinder + area MOI	✓ covered
2024	Q7	A3/A4	frustum + impulse + work	✓ covered
2024	Q8	A4	impact + accelerating car	✓ covered

Year	Full Q	Study set	Topic family	Status
2025	Q1	B1	FBD + smooth cylinder	✓ covered
2025	Q2	B2	beam + cable	✓ covered
2025	Q3	B3/B2/B4	parabolic + truss + pulley	✓ covered
2025	Q4	B4	friction + slide-tip + wedge	✓ covered
2025	Q5	A1	COM/sector/composite	✓ covered
2025	Q6	A4/A2	ore car + area MOI	✓ covered
2025	Q7	A3/A4	sphere + mallet + jet	✓ covered
2025	Q8	A3/A4	product/polar + accelerating car	✓ covered

What is not invented

The supplied regular-QB bundle does not contain a 2020 regular paper and does not contain a 2022 regular paper. They are therefore not fabricated. The 2023 backlog paper is a different exam and is intentionally excluded from this regular-QB solution.

Completeness Rule Used

When exactly the same theory/proof appears in multiple years, the full answer appears once in its set and every year is indexed above. When a numerical figure changes, a year-specific calculation/result is included when the scan supports a unique interpretation. For a small number of old beam figures whose applied-couple arrowhead is visibly soft, the full equilibrium equation is retained with the sign tied to the original arrow rather than inventing a direction.

Final 3-out-of-4 Set Revision Plan

Exam-focused finish.

Study section	Safest first	Second	Third	Backup 4th
A	A1 Centroid	A2 Area MOI	A4 Dynamics	A3 Product/Mass/Pappus
B	B1 FBD/Equilibrium	B2 Beam/Cable/Pulley	B4 Friction/Wedge	B3 Truss

48-hour high-yield checklist

- ☐ Centroid triangle + sector derivation
- ☐ Signed composite centroid table
- ☐ $J = I_x + I_y$ proof
- ☐ Parallel-axis theorem
- ☐ Composite MOI routine
- ☐ Product-of-inertia transfer/principal formula
- ☐ Pappus two theorems
- ☐ Sphere/cylinder/rod mass MOI
- ☐ $W_{net} = \Delta KE$ proof
- ☐ $R\Delta t = \Delta mv$ proof
- ☐ FBD definitions + support reactions
- ☐ Wheel-over-block minimum force
- ☐ UDL/UVL equivalent loads
- ☐ One full beam section-cut numerical
- ☐ Catenary vs parabola
- ☐ Pulley segment counting
- ☐ Truss joints + sections
- ☐ Friction angle/repose
- ☐ Slide-vs-tip threshold comparison
- ☐ Belt friction proof + wedge friction angle

Answer-writing rule

For every numerical: **Given** → **FBD/sketch** → **governing equation** → **substitution** → **unit** → **boxed final answer**. For every derivation: **assumption** → **differential relation** → **integrate** → **boundary/centroid condition** → **final boxed form**.

Prepared exclusively from the supplied CE 1201 class notes, lecture slides, regular question-bank papers and supplied solution references, with transparent independent verification where calculations were required.

Made by 2223033